

Exp. No.: 6	Define a Virtual Fence for the Robot
Date:	

Aim:

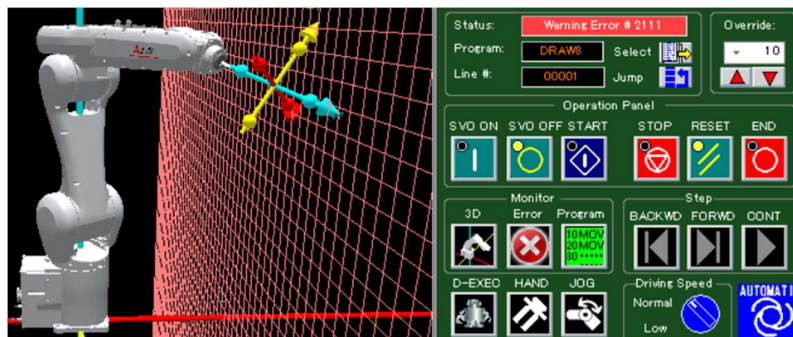
To invoke options for enabling a virtual fence around the robot to prevent it from passing a defined plane.

Procedure:

- Create a new project, > select the robot's name>Series CR800-D > 6 axis > 8 kg load > Model: RV-8CRL-D > Finish
- Select default settings while creating new workspace for this experiment.
- Select Simulator if Offline.
- Define a home position phome at (470, 0, 855, -180, 0, 180) with no flags FLG1 or FLG2 or jhome at (0, 0, 90, 0, 90, 0) for the robot.
- Locate Free Plane Limit in Robot name>Simulation>Parameter>Movement Parameter > Free Plane Limit (double click).
- From the dropdown menu select Free Plane Limit 1> Attribute ON (Inside) > Position 1 (500,0,0) > Position 2 (500, 100, 0) > Position 3 (500,100,100)>write.
- Show the plane by selecting Robot name in Layout Tree and selecting Free Plane Limit under Properties > expand Free Plane Limit > Display [] > Display [plane number].
- Write a program to pass through the plane and ensure that the robot will not pass through the plane.
- Define a Hand and attach a tool to the robot's end effector. Show that the program will stop due to the error when the tool enters the defined plane.
- Send the robot back to its home position once the drawing operation is finished.

Observation:

- The controller gives an error # 2111 when even a free plane overrun is detected either by an end effector or the attached tool.



- △  amrita
 -  3D Monitor
 - ▷  MELFA-Works
 - △  RC1
 - △  Simulation
 -  RV-8CRL-D
 -  Operation Panel
 - ▷  Program
 -  Spline
 - △  Parameter
 -  Parameter List
 - △  Movement Parameter
 -  Motion Limit
 -  Jog
 - ▷  Hand
 -  Weight and Size
 -  Tool
 -  User-defined Area
 -  Free Plane Limit
 -  Home Position
 -  Robot Additional Axis
 -  Collision Detection
 -  Warm-up Operation
 -  Movement
 -  Work Coordinate
 - ▷  Force Control
 - ▷  Collision Avoidance
 - ▷  Additional Axis Cooperative Co
 -  Program Parameter
 -  Signal Parameter
 -  Communication Parameter
 -  Field Network Parameter
 - ▷  Safety Parameter

RT Toolbox3 Program:

--	--

Results & Inference:

